Radar-Enhanced Robotic Material Perception with Vision Language Models

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ABSTRACT

Robotic perception has been significantly advanced by integrating vision with additional sensing modalities such as acoustic and tactile sensors. However, existing methods largely emphasize external object properties, including appearance and geometry, while neglecting internal material attributes (e.g., composition) that are crucial for robust and reliable robotic manipulation. In this work, we propose augmenting robotic systems with radar sensing and introduce CRMaterial, a new camera–radar fusion framework powered by vision language models (VLMs) for accurate object material identification. Preliminary experiments demonstrate that our system improves material identification accuracy by 2.5× compared to a camera-only baseline.

CCS CONCEPTS

• **Computer systems organization → Embedded and cyber-physical systems**.

KEYWORDS

Robot, vision-language models, camera-radar fusion, material identification.

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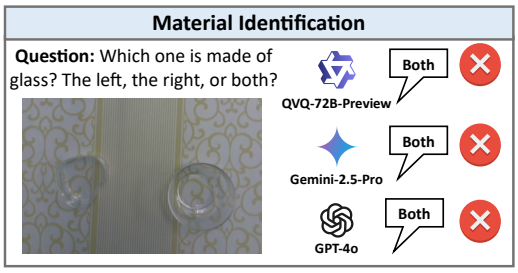


Figure 1: VLMs misclassify visually similar materials, such as a plastic box (left) and a glass bowl (right).

1 INTRODUCTION

Domestic service robots frequently carry out tasks involving object search and manipulation, where accurate material perception is crucial for safety and robustness. For instance, if a child asks the robot to pour a cup of water, it should prioritize selecting a durable plastic cup over a fragile glass one to reduce the risk of injury or breakage in the event the cup is accidentally dropped. While modern visual solutions, particularly recent advances of VLMs, demonstrate strong capabilities in recognizing objects across diverse environments, they face inherent limitations in distinguishing visually similar materials, such as transparent glass and plastic. Our preliminary test shown in Figure 1 highlights one of the key limitations of VLMs in these scenarios.

Multi-modal sensor fusion is a widely adopted approach to achieve robust robotic perception [3]. However, commonly used sensors like acoustic and tactile devices primarily capture external object characteristics such as appearance or geometry, often neglecting critical internal material properties [2]. Radar technologies offer a promising sensing modality for enhancing robotic material perception, as they provide robustness to lighting variations and exhibit strong penetration capabilities. In our prior work [1], we introduced FuseGrasp, a camera–radar fusion system designed to enable reliable grasping of transparent objects. FuseGrasp also demonstrated preliminary material identification capabilities, such as distinguishing between transparent glass and plastic cups. However, this approach is limited to closed-set object categories and depends on task-specific data collection for training deep learning models, which may restrict its scalability and practical deployment.

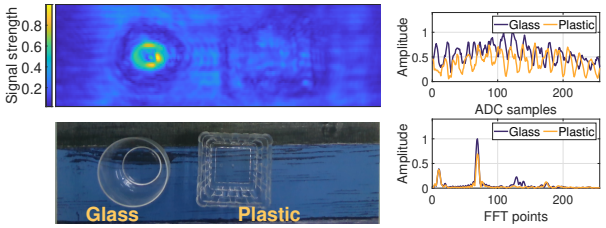
Building upon our previous work [1], this work enhances robotic material perception by introducing a VLM-based camera–radar fusion framework. VLMs have shown remarkable capabilities in vision-based object and material recognition through their rich semantic understanding. Leveraging these strengths, we propose CRMaterial, a VLM-driven camera–radar fusion pipeline that infers material composition directly from multimodal sensor data. This approach significantly improves robotic perception, with a particular focus on material identification. CRMaterial makes two key contributions. First, it leverages the precise motion control of a robotic arm to construct a synthetic antenna array, enabling the generation of high-resolution radar images that effectively capture intrinsic material properties. Second, we employ a retrieval-augmented generation (RAG) strategy to equip VLMs with domain-specific knowledge, enabling integrated reasoning over radar and RGB images. This integration allows VLMs to perform progressive reasoning across RF and optical features for open-set material identification. Preliminary results demonstrate CRMaterial's ability to distinguish multiple common materials, highlighting its potential for practical material recognition applications.

2 SYSTEM DESIGN

CRMaterial aims to achieve robust material identification in robotic systems through the integration of camera-radar fusion and VLM-based reasoning, thereby enhancing robotic perception. CRMaterial first captures radar images and RGB images, which are used as inputs to VLMs for environment understanding. Subsequently, we estimate the target's material-related electromagnetic features from the radar images. These features are provided as textual input to the VLM. Finally, we augment VLMs with a radar-related local knowledge base, employing RAG for reasoning.

Radar Imaging and Material-Related Features. In a mmWave radar system, the received signal can be approximated as $r(t) = g_r s(t-\tau)$, where $s(t)$ represents the emitted signal and τ denotes the time delay corresponding to the target's distance. The backscattering coefficient g_r characterizes the reflectivity of a point target, influenced by surface roughness and the dielectric constant of the material. To capture echo data over a larger area, we utilize the controlled movement of a robotic arm to simulate a large antenna array, enabling the synthesis of synthetic aperture radar (SAR) images. SAR imaging not only determines the precise coordinates of the target but also provides geometric information about the detected object. For SAR imaging mathematical details, see [1].

In addition to g_r, we calculate other material-related features, including the Fresnel reflection coefficient Γ_f and the relative permittivity ε_r. A corner reflector serves as an ideal



(a) Radar and RGB images. (b) Signal strengths.

Figure 2: Material features of two different objects [1].

point target, and its SAR image is used as a reference signal. Unless stated otherwise, the three parameters (g_r, Γ_f and ε_r) are normalized based on the reference target. The Fresnel reflection coefficient Γ_f quantifies the intensity of electromagnetic wave reflection at the interface between two materials. It depends on the incident angle θ, the refractive index of the material, and the polarization state of the wave. In our setup, the target is positioned directly in front of the radar i.e. $\theta = 0$, allowing Γ_f to be computed as $\Gamma_f = \sqrt{g_r}$. Relative permittivity ε_r is a complex-valued parameter that characterizes a material's capacity to polarize electromagnetic waves. It is influenced by the material's chemical composition, ambient temperature and humidity, and the operating frequency. Unless specified otherwise, references to ε_r below pertain to its real part. When $\theta = 0$, ε_r can be expressed as $\sqrt{\varepsilon_r} = (1+\Gamma_f)/(1-\Gamma_f)$. Figure 2 shows the material features are different for two transparent objects.

RAG-Enhanced VLMs. Due to the lack of radar-related knowledge in VLMs, we propose a VLM augmented with a domain-specific knowledge database. Our framework interprets radar and camera images using three key electromagnetic features includes g_r, Γ_f, and ε_r to predict the material composition of the target object. Specifically, we employ the RAG technique to enhance the VLMs' ability to identify materials. The RAG process establishes a foundation by integrating the VLM with a domain-specific knowledge base via a structured workflow. Initially, radar-related documents are embedded and indexed in a structured data repository. Once the repository is prepared, CRMaterial executes a semantic search to retrieve the most relevant document chunks. The VLM then produces an accurate response based on the retrieved information. The entire reasoning process is transparent and verifiable, as the output can be traced back to specific source documents. Consequently, CRMaterial not only provides correct recognition results but also ensures their trustworthiness through traceable evidence.

CRMaterial Prototype Implementation. We implement CRMaterial using a TI IWR6843ISK mmWave radar, which operating at a center frequency of 61.8 GHz and a bandwidth of 3.6 GHz. The mmWave radar unit is mounted alongside an Intel Realsense D435i camera on a 3D-printed stand.

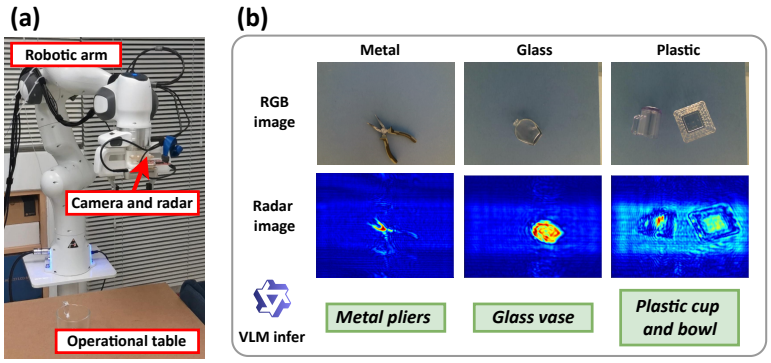


Figure 3: Preliminary Results.

Table 1: Preliminary ablation studies of CRMaterial. A total of 10 groups of data were used in the experiment.

Methods	Materials			Total
	Metal	Glass	Plastic	
RGB	2/2	1/4	1/4	4/10
RGB + radar images + $(g_r, \Gamma_f, \varepsilon_r)$	2/2	2/4	1/4	5/10
RGB + $(g_r, \Gamma_f, \varepsilon_r)$ + RAG	2/2	3/4	4/4	9/10
RGB + radar images + RAG	2/2	3/4	3/4	8/10
CRMaterial	2/2	4/4	4/4	10/10

A 7-DoF Franka robotic arm is employed to maneuver the mmWave radar and manipulation. Within our framework, Qwen team visual reasoning model, QVQ-72B-Preview, was employed for VLMs reasoning on Cherry Studio platform.

3 RESULTS AND DISCUSSION

Experimental results: To evaluate CRMaterial's performance, we selected three common materials (metal, glass, and plastic), and placed them on the operational table. The experimental setup is illustrated in Figure 3(a). The robotic arm starts from a height of 0.6m to capture an RGB image, then systematically sweeps across the table to generate a 161×84 virtual antenna array for radar imaging. Representative results including radar images, RGB images and VLM inferences are illustrated in Figure 3(b). RGB and radar images provide critical semantic context, enabling robust VLM reasoning. For instance, when a pair of pliers is placed on the table, VLM can identify the object and infers the most likely material (metal) based on semantic cues. The strong radar reflection region in the pliers' radar image further reinforces this inference during reasoning. All image and material-related textual information are processed by the RAG-enhanced VLM to execute a reasoning workflow. We tested four groups of glass and plastic samples and two groups of metal samples. The overall results are detailed in Table 1 (last row), where CRMaterial achieves 100% accuracy, demonstrating the feasibility and effectiveness of VLM-based camera-radar fusion for enhancing robotic perception.

Analysis: Preliminary ablation experiments were conducted to assess the efficacy of CRMaterial. Specifically, we evaluated the impacts of RAG module and three input modalities: RGB images, radar images, and material features $(g_r, \Gamma_f, \varepsilon_r)$. Radar data are critical for material recognition, as they provide complementary information for distinguishing visually similar plastics and glass items. When relying solely on the camera modality, the system achieves only 40% accuracy (Table 1, Row 1). Removing the RAG module significantly reduces performance, with average accuracy dropping to 50% (Row 2). This highlights that the RAG module is not only beneficial but indispensable, as it equips VLM with domain-specific knowledge for accurate radar signal interpretation. In contrast, radar images and material features $(g_r, \Gamma_f, \varepsilon_r)$ have limited impact: CRMaterial retains 90% accuracy using only material features without radar images, as these compact representations partially compensate for missing radar data (Row 3). Similarly, even without $(g_r, \Gamma_f, \varepsilon_r)$, CRMaterial performs reasoning using radar images alone (Row 4). However, for complex scenarios involving composite materials, both modalities remain essential, as radar images capture macro-scale features while $(g_r, \Gamma_f, \varepsilon_r)$ encode micro-scale material properties.

Discussion: This paper introduces CRMaterial, a robotic perception framework that strategically leverages VLM for camera-radar fusion in material identification. Our preliminary experiments demonstrate that CRMaterial achieves accurate material recognition, confirming the capability of VLM to effectively interpret material properties from radar and RGB images. We acknowledge several limitations in the current study: it serves as a preliminary validation, focusing on only three material types with limited sample sizes. Future research should pursue two key directions: (1) expanding the diversity of evaluated materials to assess generalizability, and (2) enhancing the RAG knowledge base and incorporating advanced chain-of-thought reasoning strategies to improve interpretability and accuracy.

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